

Education

Northeastern University, Boston, MA | Khoury College of Computer Sciences
B.S. in Computer Science and Game Development
GPA: 3.7 / 4.00 | Distinguished Scholar

Sep. 2022 - May 2026

Technical Skills

Programming Languages: TypeScript | JavaScript | Java | Python | C++ | C | SQL | Kotlin | Assembly | Tailwind
Frameworks & Tools: React | Node.js | Express | Vite | Next.js | Socket.io | Supabase | Flutter | Git/GitLab CI/CD | Vitest

Experience

Software Engineer | Sphere IoT

Jan. 2025 - Jun. 2025

- Developed and maintained Python services for a remote IoT virtualization lab of 100 networked devices, including an infrared command handler controlling TVs, streaming sticks, and smart-home devices.
- Collaborated with a hardware engineer to design MQTT message formats between a Python client and embedded IR controller, tracing message flow and configuration issues to ensure reliable device state changes.
- Diagnosed networking and routing failures by inspecting traffic across OSI layers and reconfiguring static IP/DHCP settings to restore device connectivity.
- Wrote automated tests within a production GitLab CI/CD workflow, reducing regressions across networked services.

Frontend Developer | Scout

Jan. 2025 - Mar. 2025

- Developed the website for the club's annual conference (300+ attendees), ensuring accessibility compliance for users with disabilities.
 - Implemented accessibility features such as color adjustments for color blindness, text scalability, and keyboard-only navigation.
 - Collaborated with a cross-functional team to deliver an inclusive user experience for a diverse audience.
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Projects

Pulse | AI-Powered CRM Insight Platform

May. 2026 - Current

- Built a serverless AWS pipeline (Lambda, DynamoDB, S3, API Gateway, CloudFront) with Terraform, provisioning 50+ resources with per-function IAM roles
- Designed an event-driven pipeline where S3 uploads trigger Lambda to parse CSVs, compute risk signals, and batch-score accounts via the Claude API with structured outputs
- Built a React/TypeScript dashboard with accessibility-tested data visualization (deuteranopia-safe color encoding via OKLab contrast validation)

Trackr | React, TypeScript, Vite, Dexie/IndexedDB, Supabase

Mar. 2026 - May 2026

- Built a modular local-first tracking app where users define and track anything (habits, moods, reps) from a single flexible schema; works fully offline via IndexedDB, then syncs to the cloud.
- Implemented cloud sync with Supabase and row-level-security auth, plus calendar and grid views, customizable per-tracker theming, and reusable templates.

GameNite | React, TypeScript, Express, Socket.io, Vitest

Mar. 2026 - May. 2026

- Collaborated in a 4-person agile team to ship 3 features to an existing web platform; led frontend across all three.
 - Built a multiplayer Mafia social deduction game, implementing phase-based UI rendering, role-specific action controls, spectator mode, and win/loss logic using React and WebSockets.
 - Wrote automated tests across server and client using Vitest and React Testing Library.
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Additional Information

Certificate: Network Fundamentals Akamai

Spoken Languages: English (fluent), Korean (fluent), Japanese (conversational)